Convolutions of Distributions

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Introduction

The distribution of a sum of independent random variables is their **convolution**. Many standard distribution families are closed under convolution: sums of independent normals are normal, sums of independent Poissons are Poisson, sums of independent gammas (with the same rate) are gamma. This closure is what makes convolution a core operation in probability.

Prerequisites

Joint distributions, independence, integration.

Theory

For independent X and Y with densities f_X and f_Y, the density of $Z = X + Y$ is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx.$$

The integral is the **convolution** $f_X * f_Y$. For discrete PMFs, the sum replaces the integral.

Characteristic functions convert convolutions to products: $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$. This makes convolutions of many distributions analytically tractable via Fourier-analytic methods.

Closure examples:

- $\mathcal{N}(\mu_1, \sigma_1^2) + \mathcal{N}(\mu_2, \sigma_2^2) \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ (independent).
- $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) \sim \text{Poisson}(\lambda_1 + \lambda_2)$ (independent).
- $\text{Gamma}(\alpha_1, \beta) + \text{Gamma}(\alpha_2, \beta) \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$ (same rate; independent).
- $\chi^2_{n_1} + \chi^2_{n_2} \sim \chi^2_{n_1+n_2}$ (independent).
- $\text{Binomial}(n_1, p) + \text{Binomial}(n_2, p) \sim \text{Binomial}(n_1 + n_2, p)$ (same p; independent).

Note that the Cauchy family is also closed under sums: sums of independent Cauchys are Cauchy – a consequence of the stable distribution family to which Cauchy belongs.

Assumptions

Independence is essential; the convolution formula fails for dependent X and Y.

R Implementation

```
set.seed(2026)

# Normal + normal is normal
X <- rnorm(1e5, mean = 0, sd = 1)
Y <- rnorm(1e5, mean = 0, sd = 1)
Z <- X + Y
c(emp_mean = mean(Z), emp_sd = sd(Z),
```

```r
  theoretical_mean = 0, theoretical_sd = sqrt(2))

# Poisson + Poisson is Poisson
X_pois <- rpois(1e5, lambda = 3)
Y_pois <- rpois(1e5, lambda = 2)
Z_pois <- X_pois + Y_pois
c(emp_mean = mean(Z_pois),
  emp_var  = var(Z_pois),
  theoretical_mean = 5,
  theoretical_var  = 5)

# Numerical convolution of two PDFs via integrate()
# f_X: uniform[0,1], f_Y: exponential(rate=1)
# Compute density of Z = X + Y at z = 0.5
f_convolve <- function(z) {
  integrate(function(x) dunif(x) * dexp(z - x, rate = 1),
            lower = 0, upper = min(1, z))$value
}
f_convolve(0.5)
f_convolve(1.5)
```

Output & Results

```
 emp_mean    emp_sd  theoretical_mean  theoretical_sd
   0.006     1.414             0.000           1.414

  emp_mean   emp_var  theoretical_mean  theoretical_var
     5.002     5.014             5.000            5.000

[1] 0.3935    # density at z = 0.5
[1] 0.3894    # density at z = 1.5
```

Empirical sums match theoretical distributions for both the continuous (normal) and discrete (Poisson) closure cases; numerical convolution computes the sum density for a non-standard case.

Interpretation

Convolutions underlie every sum-of-variables calculation in probability. The Gaussian CLT, at its core, says that sums of iid variables with finite variance converge (after standardisation) to the normal regardless of the component distributions.

Practical Tips

- For explicit convolutions of standard distributions, use the characteristic-function approach: sums of independent stable distributions are again stable.
- Numerical convolutions can be computed via FFT (`fft` in R) for efficiency on large grids.
- In discrete cases, direct summation often suffices: `outer()` followed by appropriate tabulation.
- Convolutions preserve location, scale, shape of stable families but generally destroy non-stable structure (skewness, kurtosis drift across convolutions).
- For CLT-like arguments, the speed of convergence to normal depends on the finite third moment (Berry-Esseen inequality).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots